

# Dinh Duy Kha

Postdoctoral Researcher @ University of British Columbia, Canada

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## INTRODUCTION

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I am a postdoctoral researcher at the University of British Columbia with a Ph.D. from Sungkyunkwan University. I build practical and principled defenses at the intersection of operating systems and security: memory isolation, side-channel mitigations, confidential computing, policy compliance and agentic execution. My work has been recognized with the ACM CCS Distinguished Paper Award (2023) and published at top security venues including IEEE S&P and USENIX Security.

## EDUCATION

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SUNGKYUNKWAN UNIVERSITY

Suwon, South Korea

*Ph.D. in Computer Science*

2019 - 2026

- Thesis title: “Retrofitting Operating System Security for Modern Hardware Protections”
- Synopsis: Argues that protection established by hardware and OS security must be coherent to close protection gaps exploited by modern attackers. Presents two hardware-OS coherent designs: CAPACITY [3], which enforces capability-based intra-process memory isolation via ARM Pointer Authentication and Memory Tagging, and IncognitOS [2], which redesigns OS scheduling and memory management to suppress side-channel leakage in confidential VMs.

HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Ho Chi Minh City, Vietnam

*B.S. in Computer Science*

2014 - 2019

## WORK EXPERIENCE

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SYSTOPIA LAB, UNIVERSITY OF BRITISH COLUMBIA

Vancouver, BC, Canada

*Postdoctoral Researcher*

2026 - present

SYSTEM SECURITY LAB, SUNGKYUNKWAN UNIVERSITY

Suwon, South Korea

*Graduate Student Researcher*

2019 - 2026

VNG CORPORATION

Ho Chi Minh City, Vietnam

*Software Engineering Intern – TalkTV Backend*

Jun 2018 - Jun 2019

## AWARDS

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NOTEWORTHY REVIEWER RECOGNITION, USENIX SECURITY ARTIFACT EVALUATION

2025

KOREAN GOVERNMENT BK21 RESEARCH SCHOLARSHIP

2024

DISTINGUISHED PAPER AWARD, ACM CCS

2023

## PROJECTS

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CRYPTOGRAPHIC CAPABILITIES FOR PROGRAM COMPARTMENTALIZATION [3]

- Designed an in-process capability-based compartmentalization framework leveraging ARM Pointer Authentication and Memory Tagging; evaluated its security properties.
- Developed LLVM compiler instrumentation passes to enforce memory access security policies.
- Developed a kernel module to support isolation of file-related objects using the capability-based framework.

FULL-SYSTEM MEMORY OBFUSCATION FOR CONFIDENTIAL VIRTUAL MACHINES [2]

- Designed a unikernel-based execution environment providing full-system side-channel obfuscation for confidential VMs (CVMs).

- Implemented adaptive memory rerandomization in the paging subsystem via direct MMU access, adjusting rerandomization frequency based on observed VMExit rates.
- Evaluated resilience against CVM-targeted side-channel attacks and demonstrated practical performance for real-world workloads.

## LLM-ASSISTED C TO RUST TRANSLATION

- Investigated real-world practice of developers for translating C into Rust from real-world repositories
- Constructed LLM-assisted C-to-Rust translation pipeline based on C2Rust.

## SIMULATION FRAMEWORK FOR PIM-BASED CONFIDENTIAL COMPUTING [1]

- Used gem5 to develop a full-system simulation of emerging PIM hardware.
- Implemented and evaluated security features to enable PIM as an accelerator for confidential computing.

## TEACHING

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*Graduate Teaching Assistant*, Sungkyunkwan University

- ESW4010: Special Topics on System Security (Fall 2021, Spring 2022, Fall 2022)
  - Developed an automated framework to deploy CTF challenges to docker containers.
  - Adapted the CTFd framework to make it more suitable for the classroom environment.
  - The framework used to teach subsequent courses at <https://ctf.skku.edu/>.
- SWE2001: System Programming (Fall 2021)

## ACADEMIC SERVICE

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- **Artifact Evaluation Committee:** [USENIX Security '25](#), ACM CCS '25
- **Poster Program Committee:** [ACM CCS '25](#)

## OPEN SOURCE CONTRIBUTIONS

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- [UNIKRAFT](#) – implemented AMD SEV (Secure Encrypted Virtualization) support and submitted various bug fixes to the core kernel

## SKILLS

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**Languages**    Assembly, C, C++, Rust, Python, Bash

**Systems**     KVM, QEMU, Linux kernel, device drivers, LLVM/Clang, gem5, Intel SGX, AMD SEV

## PUBLICATION

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- [1] Woo, B., **Duy, K. D.**, Han, Y., Kang, B. B., Lee, H., “Pim-oram: Towards oblivious ram primitives in commodity processing-in-memory,” in *2025 IEEE Annual Computer Security Applications Conference (AC-SAC)*, Honolulu, Hawaii, USA: ACM, Dec. 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/11391727>
- [2] **Duy, K. D.**, Kim, J., Lim, H., Lee, H., “IncognitOS: A Practical Unikernel Design for Full-System Obfuscation in Confidential Virtual Machines,” in *2025 IEEE Symposium on Security and Privacy (SP)*, Los Alamitos, CA, USA: IEEE Computer Society, May 2025, pp. 4192–4209. DOI: [10.1109/SP61157.2025.00222](https://doi.ieeecomputersociety.org/10.1109/SP61157.2025.00222) [Online]. Available: <https://doi.ieeecomputersociety.org/10.1109/SP61157.2025.00222>

- [3] Cho, K., Kim, J., **Duy, K. D.**, Lim, H., Lee, H., “RustSan: Retrofitting AddressSanitizer for efficient sanitization of rust,” in *33rd USENIX Security Symposium (USENIX Security 24)*, Philadelphia, PA: USENIX Association, Aug. 2024, pp. 3729–3746, ISBN: 978-1-939133-44-1. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity24/presentation/cho-kyuwon>
- [4] **Duy, K. D.**, Cho, K., Noh, T., Lee, H., “Capacity: Cryptographically-Enforced In-Process Capabilities for Modern ARM Architectures,” in *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS ’23, Copenhagen, Denmark: Association for Computing Machinery, 2023, pp. 874–888, ISBN: 9798400700507. DOI: [10.1145/3576915.3623079](https://doi.org/10.1145/3576915.3623079) [Online]. Available: <https://doi.org/10.1145/3576915.3623079>
- [5] **Duy, K. D.**, Lee, H., “Se-pim: In-memory acceleration of data-intensive confidential computing,” *IEEE Transactions on Cloud Computing*, vol. 11, no. 3, pp. 2473–2490, 2023. DOI: [10.1109/TCC.2022.3207145](https://doi.org/10.1109/TCC.2022.3207145)
- [6] **Duy, K. D.**, Noh, T., Huh, S., Lee, H., “Confidential machine learning computation in untrusted environments: A systems security perspective,” *IEEE Access*, vol. 9, pp. 168 656–168 677, 2021. DOI: [10.1109/ACCESS.2021.3136889](https://doi.org/10.1109/ACCESS.2021.3136889)